MD. NAZMUL ISLAM

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RESEARCH INTERESTS

Veterinary epidemiology; zoonotic and infectious disease dynamics; Bayesian and quantitative epidemiological modelling; diagnostic-test evaluation and disease surveillance; dairy herd health; and One Health approaches to animal and public health challenges in South Asian and low- and middle-income country (LMIC) settings.

EDUCATION

Master of Science in Medicine April 2025 – Present
Department of Medicine, Faculty of Veterinary Science, Bangladesh Agricultural University, Mymensingh
CGPA: 3.80 / 4.00 (first semester)
Thesis: *Transmission Dynamics, Intervention Simulation, and Economic Burden of Q Fever and Neosporosis in Large-Scale Commercial Dairy Herds of Bangladesh: A Hierarchical Bayesian Approach.*

Doctor of Veterinary Medicine (DVM) January 2019 – January 2025
Faculty of Veterinary Science, Bangladesh Agricultural University, Mymensingh
CGPA: 3.487 / 4.00

PUBLICATIONS

Author name in bold. Six peer-reviewed articles published; nine manuscripts under review; first or equally contributing first author on six.

Peer-Reviewed Journal Articles

1. Kibria, A.M., **Islam, M.N.,** Alam, M.S., Adhikari, B.J., Islam, S., Rafeh, A.A., Hasan, R.S.M., Rahman, M.S., & Rahman, A.K.M.A. (2026). Estimated true prevalence and associated risk factors of bovine paratuberculosis antibodies in dairy herds in Bangladesh. *PLOS ONE.* https://doi.org/10.1371/journal.pone.0334683
   Equally contributing first author *— study design, methodology development, Bayesian latent-class prevalence estimation, data analysis, visualization, and manuscript preparation.*
2. Alam, M.S., **Islam, M.N.,** Adhikari, B.J., Islam, S., Mahmud, R.S., Rahman, M.S., Ariful Islam, M., Aktaruzzaman, M., & Rahman, A.K.M.A. (2026). Risk factors for latent and active brucellosis in dairy cattle: a Bayesian mixed-effects multinomial regression analysis in large-scale Bangladeshi herds. *Animal Diseases, 6, 26.* https://doi.org/10.1186/s44149-026-00234-w
   Contributed to sample collection, laboratory analysis, data curation, statistical analysis, Bayesian mixed-effects multinomial modelling, and manuscript drafting.
3. Mahmud Sajeeb, M.S., Alam, M.S., **Islam, M.N.,** Islam, M.M., Adhikari, B.J., Islam, S., Rahman, M.S., & Rahman, A.K.M.A. (2025). Prevalence and risk factors of bovine viral diarrhea virus antibodies in dairy herds of Bangladesh. *Veterinary Sciences, 12(8), 739.* https://doi.org/10.3390/vetsci12080739
   Equally contributing first author *— study design, field sampling, ELISA diagnostics, multivariable regression analysis, visualization, and manuscript preparation.*
4. Aktaruzzaman, M., Ariful Islam, M., Shanta, S.A., Tahomina Akter, M., Hossain, M.T., **Islam, M.N.,** Alam, M.S., & Rahman, A.K.M.A. (2026). Prevalence and cow-level risk factors for mild and severe subclinical mastitis in intensive and extensive dairy systems in Bangladesh. *Veterinary Research Communications, 50,* 447. https://doi.org/10.1007/s11259-026-11387-x
   Contributed to fieldwork, data curation, and laboratory analysis.
5. Abu Saif, M., **Islam, M.N.,** Rahman, A.K.M.A., Ferdous, M.J., Islam, R., & Bari, F.Y. (2026). Integrated immune and protein profiles predict growth in preweaning lambs independent of feeding regimen and maternal oxytocin administration. *Iraqi Journal of Veterinary Sciences, 40(2), 397–403.* https://doi.org/10.33899/ijvs.2026.168586.4671
   Contributed to study design, methodology development, and manuscript preparation.

6. Abu Saif, M., **Islam, M.N.,** Ferdous, M.J., Rahman, A.K.M.A., & Bari, F.Y. (2026). Early-life feeding and management interventions on lamb survivability. *Bulgarian Journal of Veterinary Medicine* (online first). https://doi.org/10.15547/bjvm.2026-0050
   Contributed to study design, methodology development, formal analysis and manuscript preparation.

Manuscripts Under Review — First or Equally Contributing First Author

7. **Islam, M.N.,** Ferdous, M.J., Ema, A.A., Haque, M.Z., Alam, M.S., Rahman, M.S., & Rahman, A.K.M.A. (2026). Contrasting age-related seroepidemiological signatures of Q fever and neosporosis in large-scale commercial dairy herds in Bangladesh: a paired hierarchical Bayesian catalytic analysis. *Preventive Veterinary Medicine.* **[First author]**
8. **Islam, M.N.,** Haque, M.Z., Alam, M.S., Islam, S., Adhikari, B.J., Rahman, M.S., & Rahman, A.K.M.A. (2026). Bayesian estimation of the true prevalence of Coxiella burnetii milk antibodies and associated risk factors in dairy herds in Bangladesh. *Veterinary Medicine and Science.* **[First author]**
9. Ferdous, M.J., **Islam, M.N.,** Alam, M.S., Saif, M.A., Rahman, M.A., Aktaruzzaman, M., Rahman, M.S., & Rahman, A.K.M.A. (2026). Age-stratified transmission heterogeneity of bovine brucellosis in large commercial dairy herds in Bangladesh: a hierarchical Bayesian catalytic analysis. *Epidemics.* **[Equally contributing first author]**
10. Saha, U.K., **Islam, M.N.,** Chaudhary, P., Hassan Alif, M.K., Mim, S.N., & Rahman, M.S. (2026). Exploratory longitudinal ELISA antibody responses and hematological parameters after heat-inactivated Brucella abortus vaccination in cattle. *Bulgarian Journal of Veterinary Medicine.* **[Equally contributing first author]**

Manuscripts Under Review — Co-Author

11. Alam, M.S., **Islam, M.N.,** Adhikari, B.J., Islam, S., Mahmud, R.S., Rahman, M.S., Ariful Islam, M., Aktaruzzaman, M., Meletis, L., Kostoulas, P., & Rahman, A.K.M.A. (2026). Bayesian hierarchical mixture modelling to derive probabilistic iELISA thresholds for bovine brucellosis in endemic dairy systems. *PLOS ONE.* Accepted for publication.
12. Alam, M.S., **Islam, M.N.,** Adhikari, B.J., Islam, S., Rahman, M.S., Islam, M.A., Aktaruzzaman, M., & Rahman, A.K.M.A. (2026). Economic burden and cost-effectiveness of brucellosis control strategies in large commercial dairy herds in Bangladesh: a Bayesian decision-analytic study. *Preventive Veterinary Medicine.*
13. Anwarul Islam, **Islam, M.N.,** Ferdous, M.J., Alam, M.S., Rahman, M.A., Mahmud, R.S., Alam, M.M., & Rahman, A.K.M.A. (2026). A probabilistic decision-analytic model of the economic burden and strategic control of ruminant fascioliasis in Bangladesh. *Veterinary Parasitology: Regional Studies and Reports.*
14. Aktaruzzaman, M., Ariful Islam, M., Shanta, S.A., Tahomina Akter, M., Hossain, M.T., Alam, M.S., **Islam, M.N.,** Meletis, L., Kostoulas, P., & Rahman, A.K.M.A. (2025). Bayesian Gaussian mixture modelling to determine somatic cell count thresholds for subclinical mastitis diagnosis in dairy cows. *The Journal of Veterinary Medical Science.*
15. Aktaruzzaman, M., Tahomina Akter, M., Mazhar Islam, M., Ariful Islam, M., Hossain, M.T., Alam, M.S., **Islam, M.N.,** & Rahman, A.K.M.A. (2025). Economic impact of subclinical mastitis in Bangladeshi crossbred dairy cattle. *Preventive Veterinary Medicine.*

CONFERENCE PRESENTATIONS & PROCEEDINGS

1. **Islam, M.N.,** Alam, M.S., Islam, S., Adhikari, B.J., & Rahman, A.K.M.A. (2025). Feasibility of on-farm somatic cell count screening for subclinical mastitis: prevalence and risk factors in a commercial dairy herd in Bangladesh. *Oral presentation,* 1st International Online Conference on Veterinary Sciences (IOCVS 2025), MDPI, Basel, Switzerland. https://sciforum.net/paper/view/27422
2. Alam, M.S., **Islam, M.N.,** Rafeh, A.A., Adhikari, B.J., Kibria, M.A., Mahmud, M.S., Ema, A.A., Islam, S., Rahman, M.S., & Rahman, A.K.M.A. (2025). Herd-level prevalence of brucellosis, paratuberculosis, neosporosis, and bovine viral diarrhea in large dairy herds in Bangladesh. *BAU Research Progress, 35.*
3. Alam, M.S., **Islam, M.N.,** Rafeh, A.A., Rahman, M.S., Islam, M.A., & Rahman, A.K.M.A. (2024). Herd-level prevalence of brucellosis in large dairy herds in Bangladesh. *Proceedings of the BSVER Annual Scientific Conference 30,* Bangladesh Agricultural University, Mymensingh, 24–25 February 2024.

4. **Islam, M.N.** (2024). Prevalence and epidemiology of bovine brucellosis in large dairy farms of Bangladesh. *Poster presentation,* BSVER Annual Scientific Conference 2024, Bangladesh Agricultural University, Mymensingh, 24–25 February 2024.

RESEARCH EXPERIENCE

Research Assistant · Epidemiology & Preventive Medicine Lab, Dept. of Medicine, BAU Jan 2023 – Present

- Developed reproducible Bayesian modelling workflows in R and Stan, including hierarchical catalytic transmission models, latent-class prevalence estimation, and mixture models for diagnostic-threshold derivation.
- Designed and conducted field and laboratory studies of zoonotic and infectious diseases in ruminants, contributing to peer-reviewed publications.
- Performed indirect ELISA, PCR, and DNA extraction for serological and molecular diagnostics.
- Co-authored manuscripts as first or equally contributing author, and produced publication-quality figures in ggplot2 and Adobe Illustrator.
- Mentored junior researchers in study design, data collection, and laboratory procedures.

TECHNICAL & RESEARCH SKILLS

Statistical & Bayesian modelling: R (rstan, brms, tidyverse, ggplot2); Stan; hierarchical Bayesian catalytic models (force of infection, R_0); latent-class and mixture models; mixed-effects and multivariable regression; diagnostic-test evaluation; prevalence and risk-factor analysis

Reproducible research: R Markdown, Git and GitHub, structured data management

Laboratory techniques: indirect ELISA, DNA extraction, PCR, microscopy, histopathology

Geospatial analysis: QGIS (spatial data visualization and disease mapping)

Scientific visualization: ggplot2, Adobe Illustrator, Adobe Photoshop, BioRender, Canva

Productivity & languages: Microsoft Office Suite; English (full professional proficiency; medium of instruction throughout academic career)

AWARDS, FELLOWSHIPS & HONORS

- **National Science and Technology (NST) Fellowship** — Ministry of Science and Technology, Government of Bangladesh (2026). Awarded in support of MS thesis research in veterinary medicine and animal health, in recognition of academic merit and research potential.
- **Aspire Leaders Program — Finalist (2025).** Selected among 6,175 finalists from over 36,000 applicants; completed 40 hours of coursework by the Aspire Institute (founded by Harvard Business School faculty).
- **Education Board Merit Scholarship** — Secondary School Certificate (2015) and Higher Secondary Certificate (2017).

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Veterinary Internship** — Faculty of Veterinary Science, BAU (Jul – Dec 2023). Clinical assessment, treatment, and surgical assistance across clinical, farm, and wildlife settings.
- R Programming for Research — Udemy (2024).
- Agricultural Extension Education (6-day field training) — Department of Agricultural Extension Education, BAU (2024).

PROFESSIONAL MEMBERSHIPS & SERVICE

- Student Member, American Society for Microbiology (Global Outreach) — Jun 2024 – Present.
- Member, Badhon (Voluntary Blood Donors' Organization) — Feb 2019 – Present.

REFERENCES

Prof. Dr. A. K. M. Anisur Rahman — Department of Medicine, Faculty of Veterinary Science, BAU
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